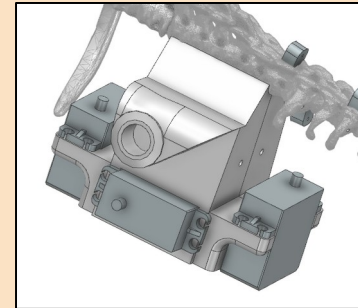
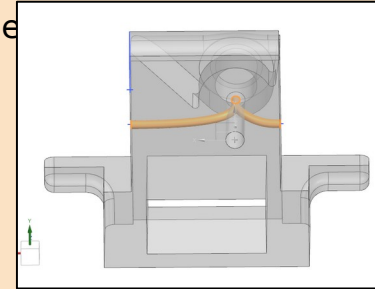
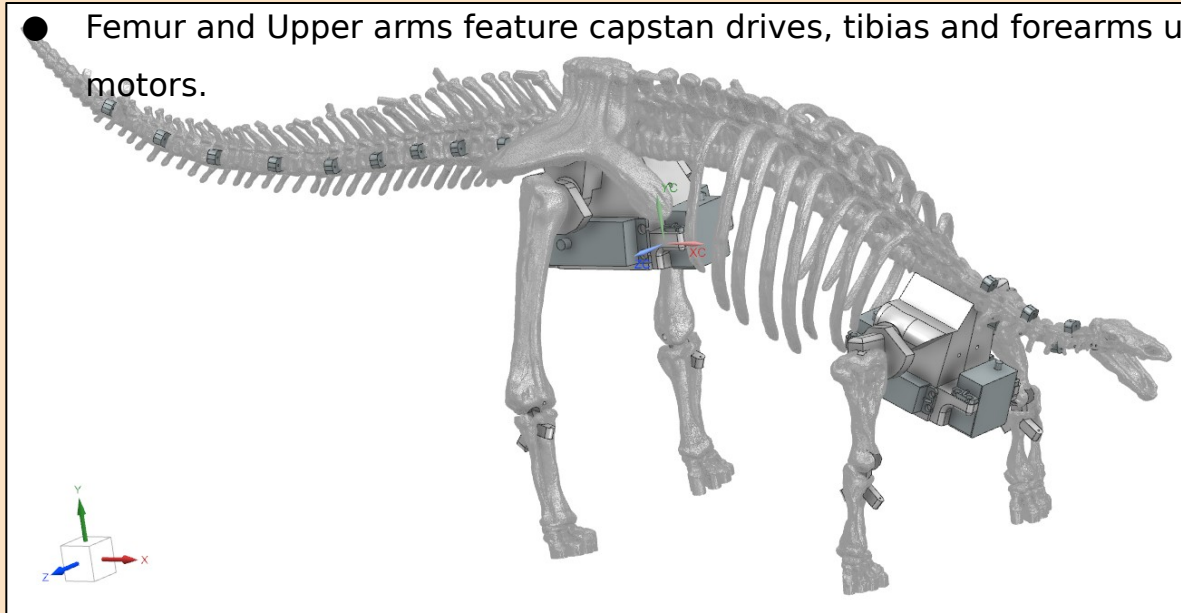


Bring ROAR to Life

- The goal of Bring ROAR to Life is to design and create an anatomically accurate Stegosaurus that is able to walk and interact with its environment in a lifelike manner
 - Our end goal of an ideal project will be to provide an educational resource for museums in general
-

3D Modeling

- Current layout with primary motor placement.
- Largely based on 3D scan data with mechanical components added to the existing structure.
- Femur and Upper arms feature capstan drives, tibias and forearms use motors.



3D Modeling

- Light FEA was run to get baseline estimates for strength.
- Based off maximum motor stall torque.
- Limb torque calculations based off gait study and maximum angular acceleration in joints

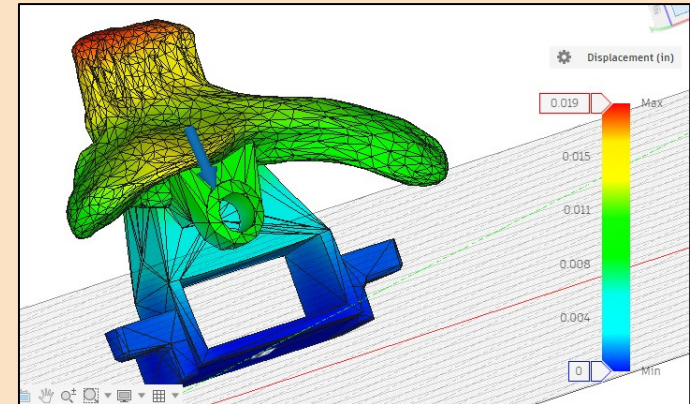
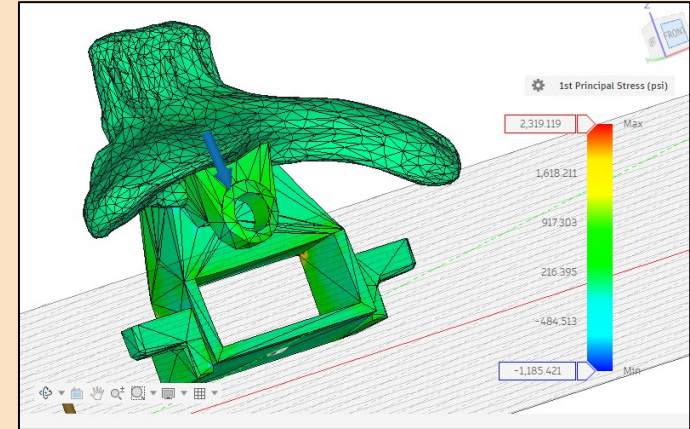


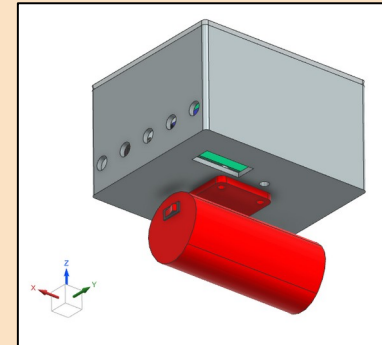
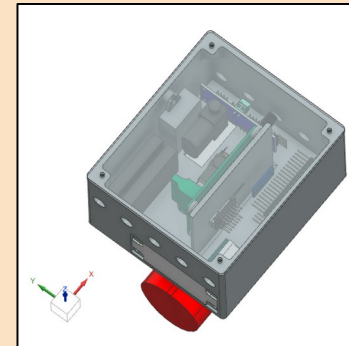
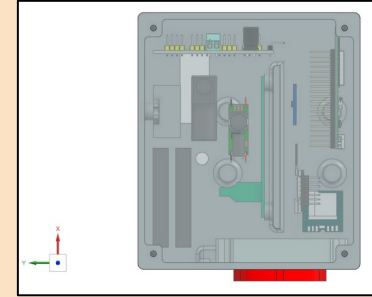
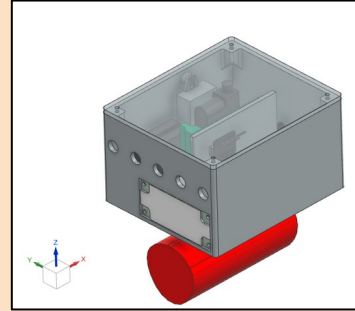
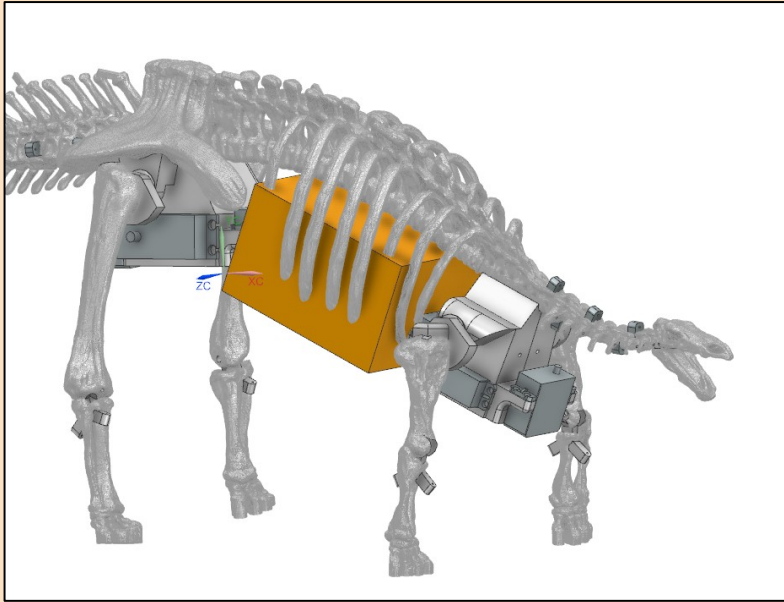
Table 3: Torque Data

	Front_Femur	Front_Tibia	Rear_Femur	Rear_Tibia
Max Absolute Angular Acceleration	8298.22	9326.6	6518.8	11216.47
Length (in)	4.0000	3.4000	7.8200	5.2800
Mass (lb)	1.19E-01	1.01E-01	2.33E-01	1.58E-01
Max Possible Torque (lb*ft)	3.06	2.11	17.954	9.51

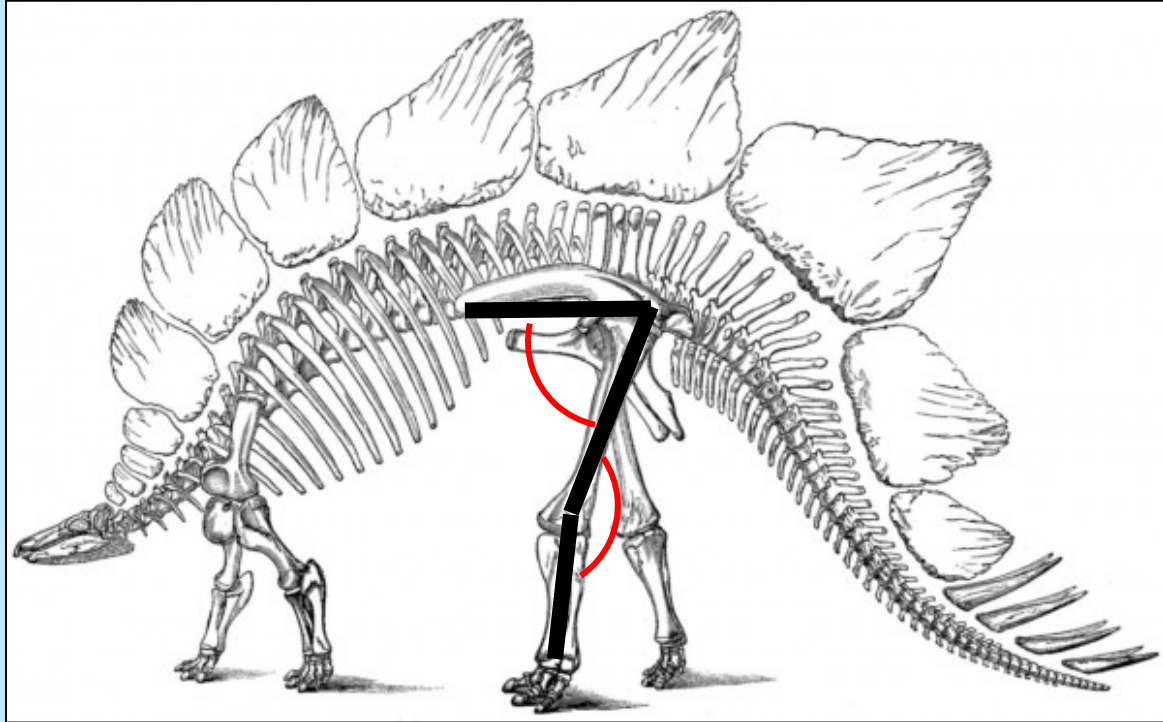
Electronics Packaging

- This housing will fit under the ribcage
- The housing for the electronics is approximately 5.2in. x 4.5in. x 3in
- Many components are fitted precisely into a tight space
- The cover will be connected to the dinosaur's spine, and the housing will be connected to the cover
- Some components will be bolted into the housing, and other components will be free floating in their designated space
- Cutouts were made so that wires can go out of the housing and into the motors

Electronics Packaging



Gait Study

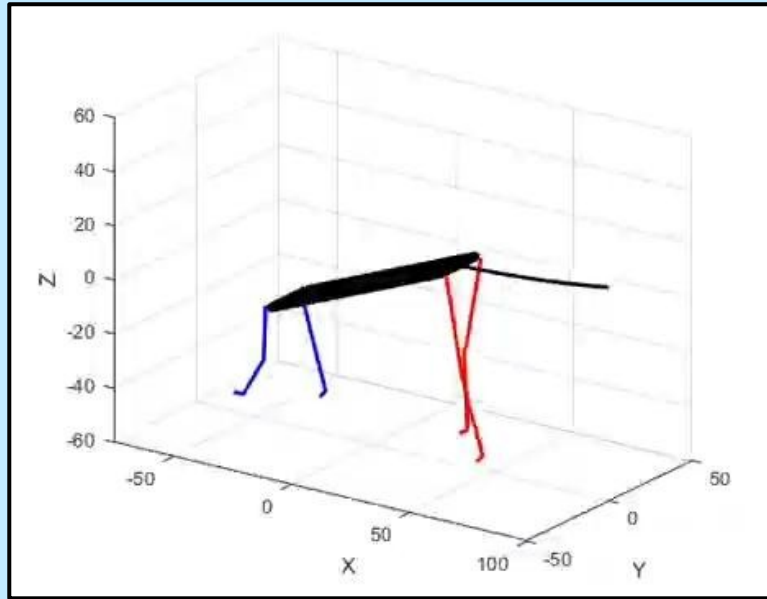


Gait Study

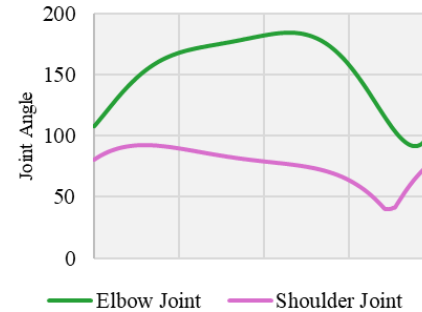
Front Leg Joint Angles			
Stride Status	Gait Percent	Hip Angle	Knee Angle
Heel Touch	0%	135°	150°
Full Load	50%	115°	210°
Toe Off	65%	105°	150°
Leg Swing_1	75%	70°	100°
Leg Swing 2	85%	120°	95°
Heel Touch	100%	135°	150°

Rear Leg Joint Angles			
Stride Status	Gait Percent	Hip Angle	Knee Angle
Heel Touch	0%	75°	160°
Full Load	50%	90°	175°
Toe Off	65%	100°	160°
Leg Swing 1	75%	110°	150°
Leg Swing 2	85%	95°	130°
Heel Touch	100%	75°	160°

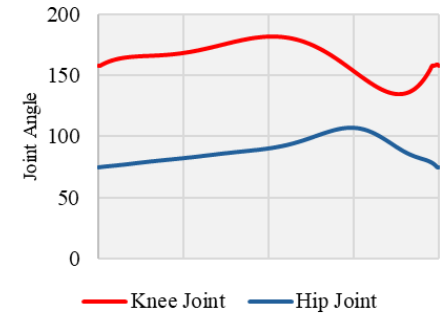
Gait Study



Front Leg Joint Angles, Rev 1

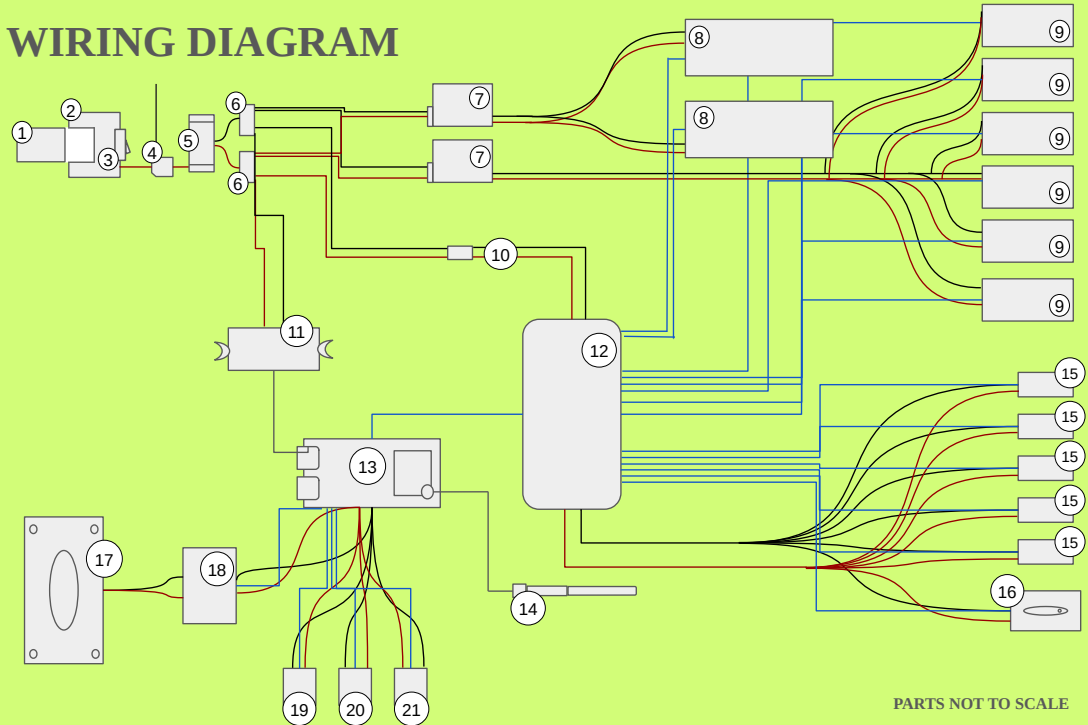


Rear Leg Joint Angles, Rev 1



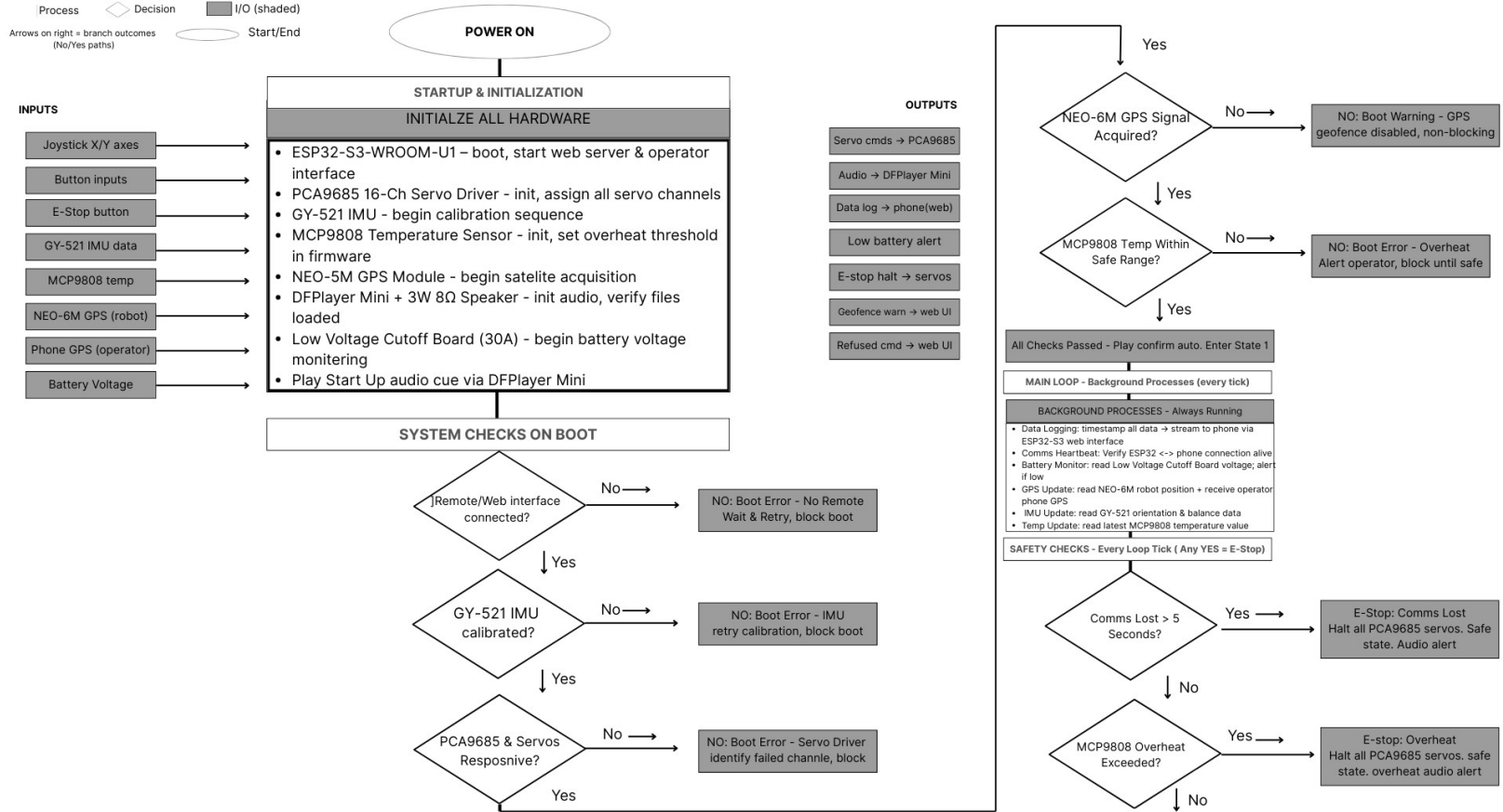
ITEM NO.	SUBSYSTEM	COMPONENT	QTY
1	Power	Milwaukee M12 Battery	1
2	Power	Battery Adapter	1
3	Power	Switch	1
4	Power	Fuse	1
5	Power	Low Voltage Cutoff Board (30A)	1
6	Power	BusBar	2
7	Motor	2-8S LiPo UBEC-8A For 5 Servos Motors	2
8	Motor	DS5180 High-Torque Servo - 80Kg servo	2
9	Motor	DS3230MG Servo - 30Kg servo	6
10	Motor	5V 3A Low Volt Buck Converter	1
11	Motor	5V 3A Buck Converter USB-C	1
12	Motor	PCA9685 16-Ch Servo Driver	1
13	Control	ESP32-S3-WROOM-U1	1
14	Control	Antenna	1
15	Motor	Micro Linear Servo	5
16	Motor	Gotek 1.5g Micro Digital Linear Servo Actuator	1
17	Audio	3W 8Ω Speaker	1
18	Audio	DFPlayer Mini MP3	1
19	Control	MCP9808 Temp Sensor	1
20	Control	GY-NEO6MV2 NEO-6M GPS Module	1
21	Control	Additional GY-521 IMUs	1

WIRING DIAGRAM

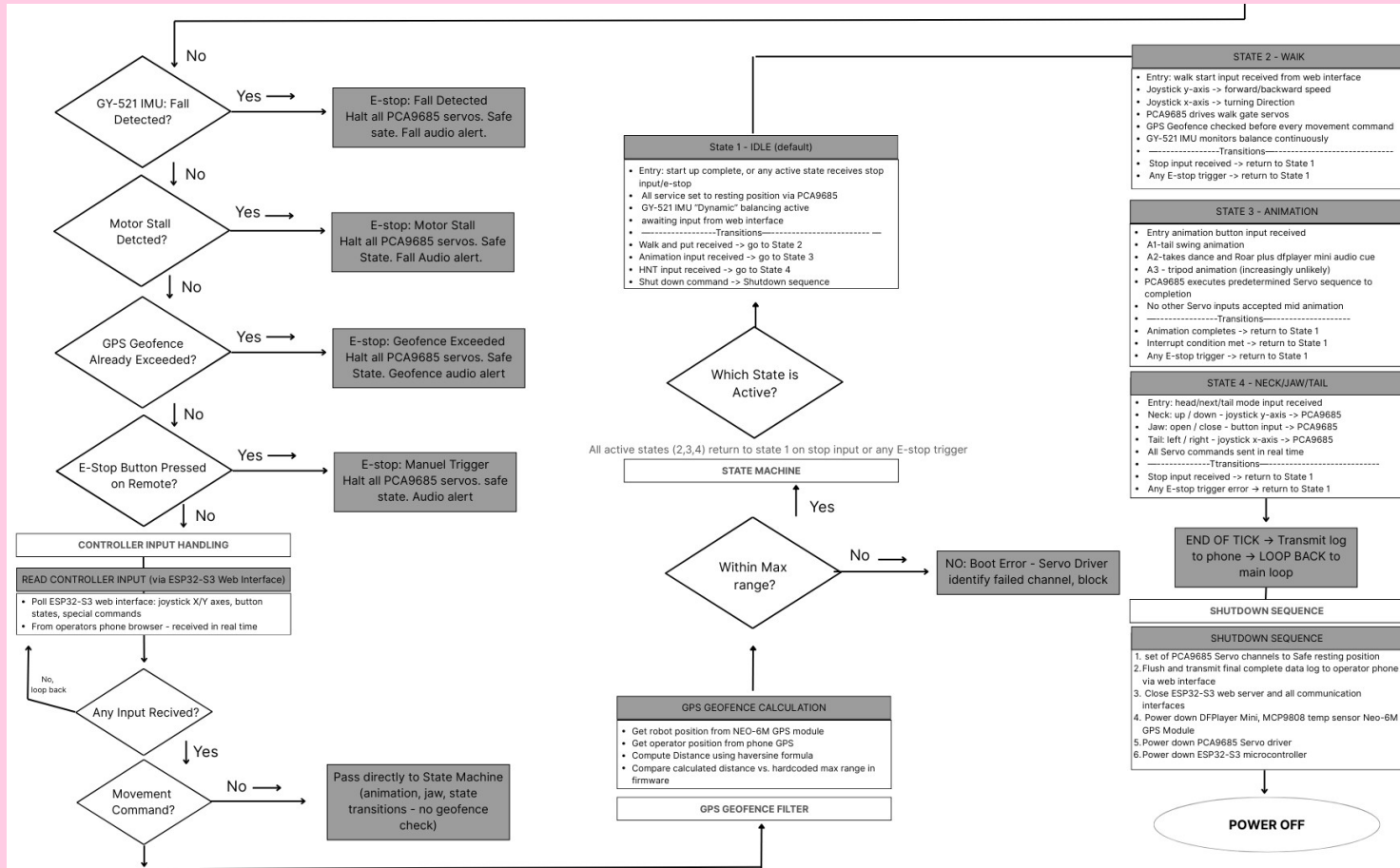


PARTS NOT TO SCALE

ROBOT CONTROL SYSTEM - Full Code Flowchart

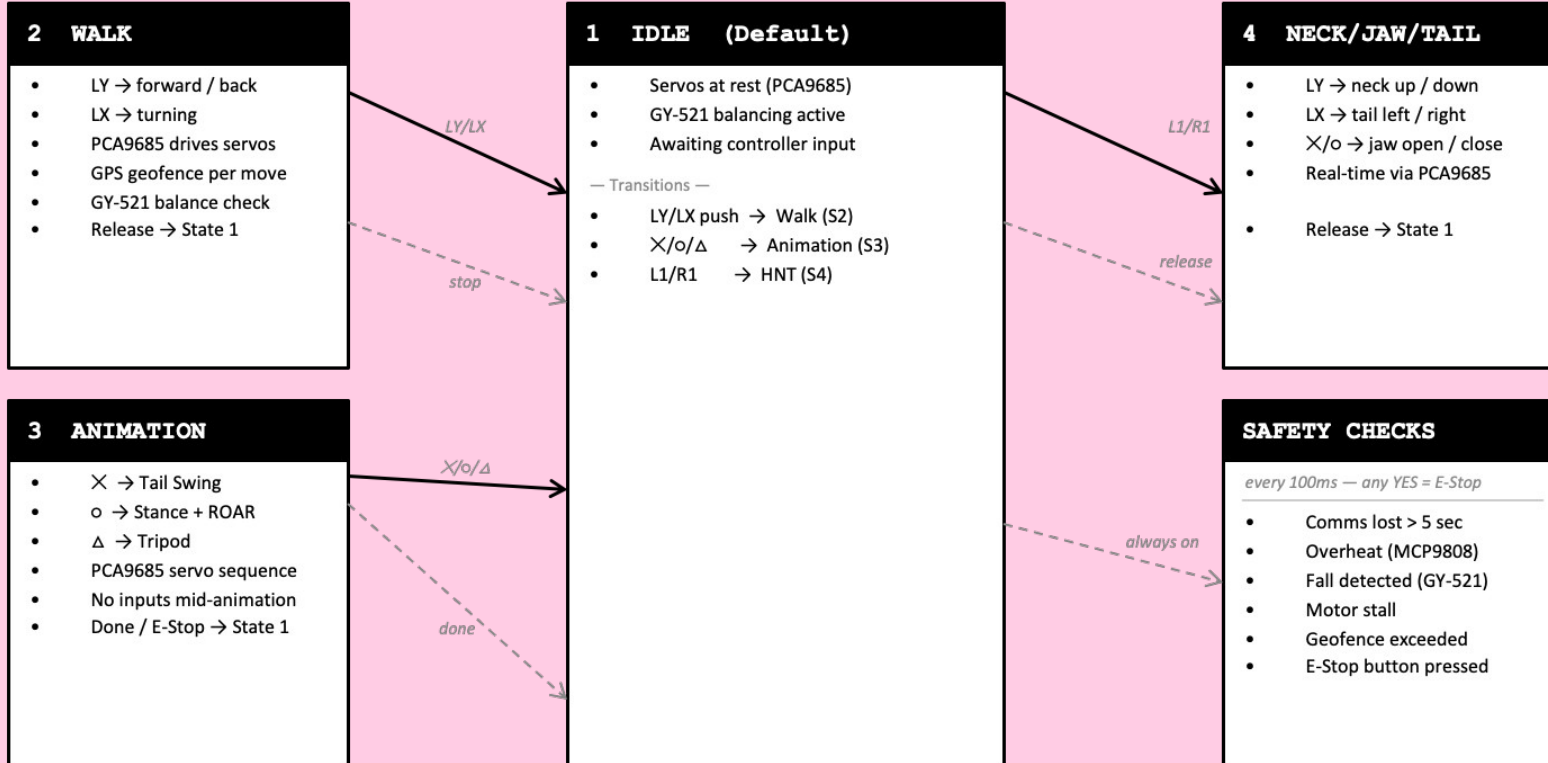


Control System



Robot Control – State Machine

All active states return to State 1 on stop input or E-Stop



Robot Controller



DISCONNECTED

State: " "

Temp: "°C

Dist: "m

GPS: " "

IDLE

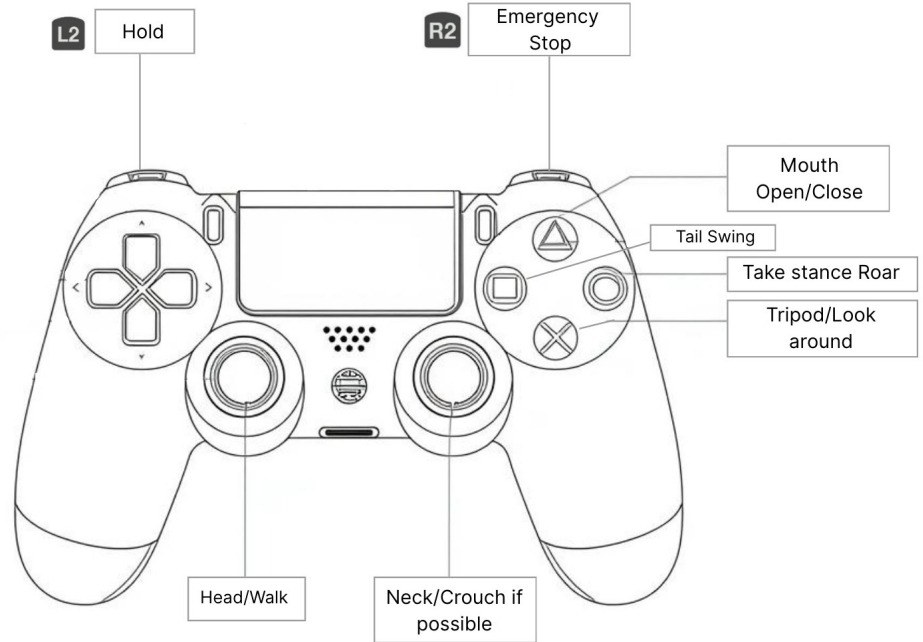
E-STOP

No controller detected " connect a PlayStation controller

LX: 0 LY: 0

Tentativ

The visual design is subject to change

Tentative Proposed Controller Layout

Finances

Approved budget:
\$530.00

Budget used: \$427.38

Amount Remaining:
\$102.62

TOTAL (with Tax):			\$427.38	
Item ID	Name	Quantity	Total Cost	Sourced From
001	Black ASA Filament	1kg	\$24.00	Personal
002	Off-White ASA Filament	1kg	\$24.00	Personal
003	White PETG Filament	1kg	\$15.00	Personal
004	1.3mm UHMWPE Cord (Black)	100ft	\$12.00	Purchase
005	black, white, grey	4kg	\$80.00	Personal
006	M12 Battery Adapter	1	\$18.00	Purchase
007	Milwaukee M12, 2.0, 4.0, and 6.0 battery	3	\$0.00	Personal
008	Mini Speakers	1	\$11.99	Purchase
009	ESP32-S3 Microcontroller	1	\$19.99	Purchase
010	DS5180 80kg servo	2	\$69.98	Purchase
011	DS3230 30kg servo (2-pack)	3	\$90.87	Purchase
012	1.5g Micro Linear servo actuator	1	\$12.53	Purchase
013	16 channel PWM Servo Driver	1	\$12.99	Purchase
014	5V 3A Buck Converter USB-C	1	\$11.99	Purchase
015	2-8S LiPo servo power supply	2	\$37.98	Purchase
016	Low Voltage Cutoff Board (30A)	1	\$19.99	Purchase
017	MCP9808 Temperature Sensor	1	\$4.41	Purchase
018	MPU 6050 Gyro./Accel. Module	1	\$6.99	Purchase
019	GY-NEO6MV2 GPS Module	1	\$19.99	Purchase
020	5V 3A Low Volt Buck Converter	1	\$9.95	Purchase
021	GNSS GPS module	1	\$19.90	Purchase
022	Mini MP3 Player	1	\$9.90	Purchase
023	Threaded Instart Bolts	1	\$9.97	Purchase

Thank you!
